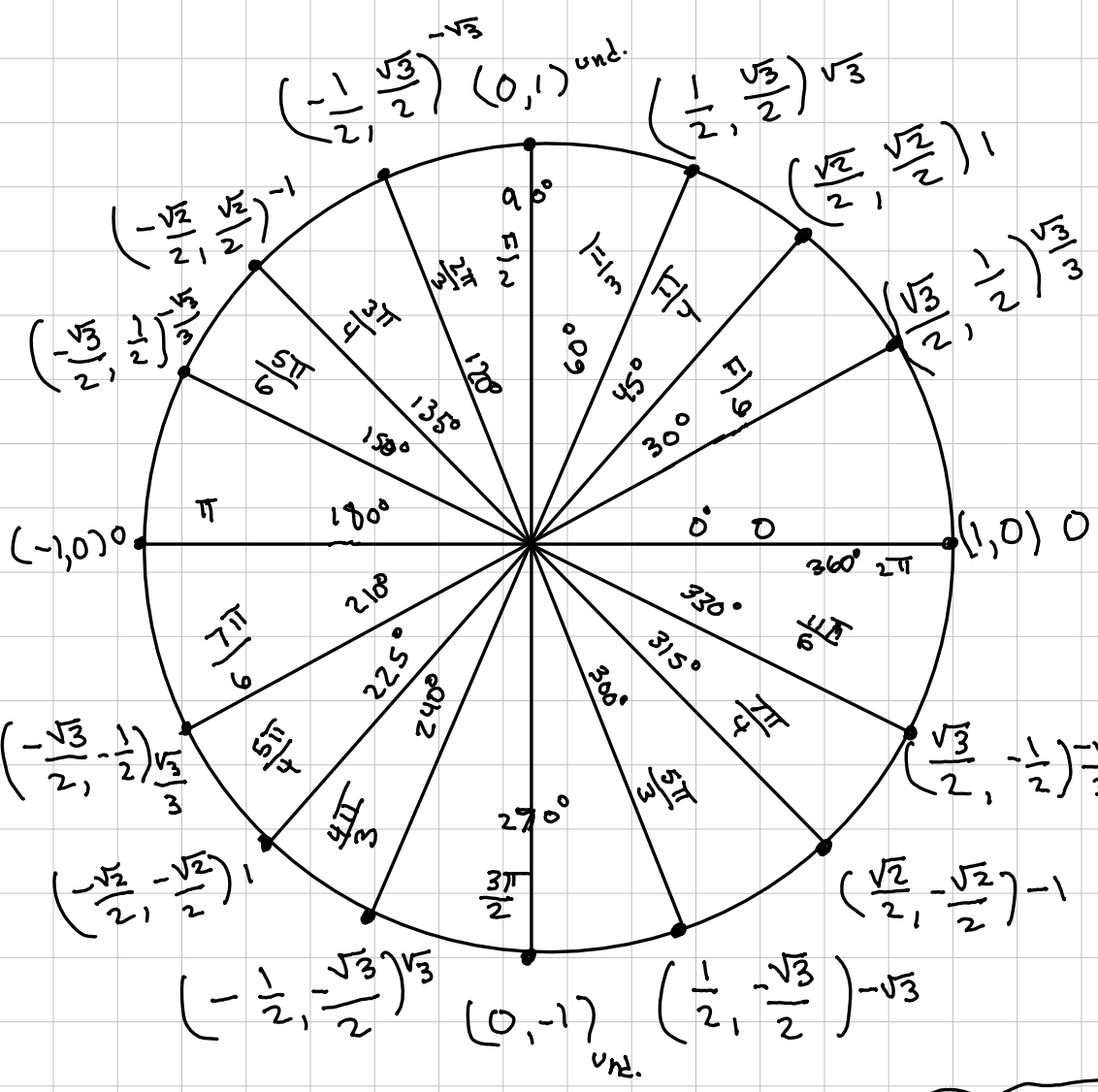


$\sin = \frac{opp}{hyp} \longrightarrow \csc = \frac{hyp}{opp}$

$\cos = \frac{adj}{hyp} \longrightarrow \sec = \frac{hyp}{adj}$

$\tan = \frac{opp}{adj} \longrightarrow \cot = \frac{adj}{opp}$

$\sin = x$

$\cos = y$

$\tan = \frac{y}{x}$

Double Angle Formulas

$\sin 2\theta = 2\sin\theta\cos\theta$

$\cos 2\theta = \cos^2\theta - \sin^2\theta$ or $2\cos^2\theta - 1$ or $1 - 2\sin^2\theta$

$\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$ or $\frac{\sin 2\theta}{\cos 2\theta}$

Law of Sine

$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$

Law of Cosine

$a^2 = b^2 + c^2 - 2bc \cdot \cos A$ | $\cos A = \frac{b^2+c^2-a^2}{2bc}$

$b^2 = a^2 + c^2 - 2ac \cdot \cos B$ | $\cos B = \frac{a^2+c^2-b^2}{2ac}$

$c^2 = a^2 + b^2 - 2ab \cdot \cos C$ | $\cos C = \frac{a^2+b^2-c^2}{2ab}$

Area of a Triangle

SAS

Area $= \frac{1}{2}bc \cdot \sin A$

Area $= \frac{1}{2}ac \cdot \sin B$

Area $= \frac{1}{2}ab \cdot \sin C$

SSS

① Find s

$s = \frac{a+b+c}{2}$

② Plug into formula

Area $= \sqrt{s(s-a)(s-b)(s-c)}$